MILITARY SPECIFICATION

SEARCHLIGHT, INFRARED 28 VOLT, 50-60 AMPERE AN/VSS-3

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers one type of xenon searchlight for projecting either visible or infrared radiation, (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	Strapping, Steel, Flat and Seals
PPP-B-20	Bags, Cotton, Mailing
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock Corner
PPP-B-636	Boxes, Shipping, Fiberboard
PPP-P-291	Paperboard, Wrapping and Cushioning
PPP-T-45	Tape, Gummed, Paper, Reinforced and Plain, For Sealing and Securing

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MIL-P-116	Preservation-Packaging, Methods of
MIL-T-152	Treatment, Moisture and Fungus-Resistant of Communications, Electronics, and Associated Electrical Equipment
MIL-S-901	Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for
MIL-C-6021	Castings, Classification and Inspection of

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MIL-M-10304	Meters, Electrical Indicating, Panel Type; Ruggedized, General Specification for
MIL-P-11268	Parts, Materials, and Processes Used in Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-T-22361	Thread Compound; Antiseize, Zinc Dust-Petrolatum
MIL-P-26514	Polyurethane Foam, Rigid or Flexible For Packaging
MIL-B-43014	Boxes: Water Resistant Paperboard; Folding, Set-Up, and Metal-Stayed.
MIL-F-45289	Filter, Infrared Interference-Type (Pink) for Searchlight, Infrared, AN/VSS-3, 28 Volt, 50-60 Ampere
MIL-R-55344	Radiation Source, Infrared, for Searchlight, Infrared, AN/VSS-3

STANDARDS

FEDERAL

FED-STD-595	Colors

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-171	Finishing of Metal and Wood Surfaces
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-252	Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired, and other Devices
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-781	Reliability Tests: Exponential Distribution
MIL-STD-810	Environmental Test Methods

DRAWINGS

ELECTRONICS COMMAND

SC-DL-646800	Searchlight, Infrared

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(Copies of specifications, standards and drawings required by suppliers in connection with specific procurements functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on the date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials

ASTM B-26	Aluminum Alloy Sand Castings
ASTM B 108	Aluminum Alloy Permanent Mold Castings

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Description. The searchlight shall consist of the control searchlight set main, the control searchlight set remote, appropriate cable assemblies, appropriate mounting kits (see 6.2) and the basic assembly. The basic assembly consists of a radiation source, interference filter, heat exchanger, ignition and operation electronics, reflector assembly, housing and a canvas cover.

3.2 Construction. The searchlight shall be constructed in accordance with SC-DL-646800 and as specified herein.

3.3 First article. When specified in the contract (see 6.2) the contractor shall furnish first article units for inspection (see 4.3).

3.4 Parts and materials. Parts and materials shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the contractor in accordance with MIL-P-11268.

3.4.1 Radiation Source Infrared (RSI). The RSI shall be in accordance with MIL-R-55344.

3.4.2 Filter. The filter shall be in accordance with MIL-F-45289.

3.5 Operation and environmental.

3.5.1 Starting and normal operation. The searchlight shall ignite and maintain ignition within 3 seconds upon the application of any voltage from 24 to 28 volts dc. The input current shall not exceed 60 amperes at 28 volts dc, in any environment specified herein.

3.5.2 Time to change operational modes. The searchlight shall operate in any environment specified herein in the compact (visible and infrared) and spread-beam (visible and infrared) modes, (see 6.3). The time required to change from one mode of operation to another shall not exceed 3 seconds.

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3.5.3 Alinement retention. The searchlight shall retain optical alinement within plus or minus 5 milliradians after experiencing the environmental conditions contained herein.

3.5.4 Light tightness. No visible light from the searchlight (except the red glow permitted through the front glass cover during operation in the infrared mode), (see 6.3), shall be detectable at any distance greater than 25 feet in any direction with the searchlight operating at 28 volts dc input and in the infrared mode.

3.5.5 Angular visual security. The red glow during operation in the infrared mode shall not be visible outside 2-1/2° each side of the optical centerline.

3.5.6 Beam characteristics. With the searchlight operating at 28 volts dc maximum and drawing 60 amperes maximum the following beam characteristics shall be obtained.

3.5.6.1 Compact mode. Peak beam luminous intensity shall be not less than 60 million candelas with a beam width of 1.0°, plus or minus 0.25° as measured between two points where the luminous intensity is 10 percent of the maximum value.*
*NOTE: A thirty percent linear degradation in peak luminous intensity and an increase in beam width not to exceed 1.5° is allowable over the specified life of the RSI (MIL-R-55344).

3.5.6.2 Spread-beam mode. Spread-beam luminous intensity shall have a beam width of 6.0°, plus 1.0° minus 0.5° as measured between two points where the luminous intensity is 10 percent of the maximum value. The maximum intensity in the spread-beam mode shall be within 1.0° of each side of the center of the beam (2.0° total).

3.5.6.3 Uniformity. The luminous intensity of both compact and spread-beam modes shall range from 10 percent of the maximum intensity point thru the maximum to the other 10 percent of the maximum intensity point without intermediate increases or decreases where the intensity differential between a valley and the nearest adjacent peak exceeds 10 percent of the maximum intensity.

3.5.7 RSI cooling. After searchlight operation for 15 minutes in the infrared mode at 28 vdc in an ambient temperature of plus 23°C, plus or minus 5°C, the air exhaust temperature shall be plus 47°C, plus or minus 5°C.

3.5.8 Reliability. The AN/VSS-3 searchlight shall have a specified Mean-Time-Between-Failure (MTBF) of 500 hours when operating under the following environmental conditions:

Temperature	Minus 15°C to plus 40°C.
Duty cycle	2 hours consisting of 1.6 hours ON, 0.4 hour OFF
Vibration	0.06 inch total excursion at any nonresonant frequency between 10 and 55 hertz (Hz) and 5 g's input from 55 to 500 Hz

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3.5.9 Vibration. The searchlight shall start and operate without damage (see 6.3) at room temperature (see 6.3) and at minus 54°C, while being subjected to sustained vibrations between 10 and 55 Hz at 0.03 inch (0.06 inch, plus or minus 10 percent total excursion) and 5 g's from 55 to 500 Hz.

3.5.10 Mechanical shock. While operating in the compact mode, the searchlight shall retain a minimum of 60 million candelas with a beam width of 1.0°, plus or minus 0.25° and shall not be damaged after being subjected to mechanical shocks of 55 g's (see 6.3.), plus or minus 5 g's for 6 milliseconds, plus or minus 0.5 millisecond duration.

3.5.11 Immersion. The searchlight shall not be damaged or leak water into the housing during immersion in fresh water for 30 minutes with the highest point of the searchlight 3 feet below the surface of the water.

3.5.12 Fungus and moisture resistance. The electrical circuitry, including all components and connections, shall be protected from effects of moisture and fungus growth by a treatment conforming to MIL-T-152, Type II. The heat exchanger and optical components (RSI, Reflector, IR Filter) shall not be sprayed.

3.5.13 Altitude. The searchlight shall operate as specified herein at 10,000 feet above sea level with plus 35°C ambient temperature.

3.5.14 Altitude transportation. The searchlight shall not be damaged by transportation in unpressurized aircraft at any altitude to 40,000 feet above sea level at minus 62°C temperature.

3.5.15 Salt fog. The searchlight shall not be damaged by exposure (in both operation and storage conditions) to salt-sea atmosphere.

3.5.16 Storage temperatures.

3.5.16.1 Low temperature storage. The searchlight shall not be damaged by storage in any ambient temperature from plus 23°C to minus 62°C.

3.5.16.2 High temperature storage. The searchlight shall not be damaged by storage in any ambient temperature from plus 23°C to plus 68°C at all possible relative humidities within that range.

3.5.17 Operating temperatures.

3.5.17.1 Low temperature operation. The searchlight shall perform as specified herein at any ambient temperature from plus 23°C to minus 62°C.

3.5.17.2 High temperature operation. The searchlight shall perform as specified herein at any ambient temperature from plus 23°C to plus 52°C at all possible relative humidities within that range.

3.5.18 Electromagnetic compatibility. The electromagnetic compatibility of the searchlight shall conform to MIL-STD-461, except during ignition, equipment class-tactical non-CE, class II C.

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3.6 Treatment and painting. Unless otherwise specified, the searchlight and its components and parts shall be cleaned, treated and painted in accordance with MIL-STD-171. Painted surfaces shall have finish number 5.2 for iron and steel, 7.4 for aluminum and 20.8 for all metals. Color shall be olive drab lusterless, number X34087 of FED-STD-595. This includes finish of hardware such as handles, hinges, screws, etc., and necessary touch-up before delivery.

3.7 Marking. Unless otherwise specified, the searchlight and all components shall be marked and identified per MIL-M-13231.

3.8 Castings. Unless otherwise specified, castings shall conform to ASTM B 26 or ASTM B 108.

3.9 Lubrication. All mating aluminum threads shall be treated with lubricant per MIL-T-22361.

3.10 Workmanship. Unless otherwise specified, workmanship shall conform to Requirement 9 of MIL-STD-454.

3.11 Soldering. Unless otherwise specified, soldering shall conform to Requirement 5 of MIL-STD-454.

3.12 Welding. Unless otherwise specified, welding shall conform to Requirement 13 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article (see 4.3).

b. Quality conformance inspection (see 4.5)

c. Inspection of preparation for delivery (see 4.7)

4.3 First article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

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4.3.1 Inspection. All materials, parts, processes and assemblies shall be examined for defects in conformance with the inspection plan in the applicable specification or on the drawing. In addition, the following inspection plans shall be applicable as listed in Table I.

TABLE I Inspection

Inspection	Requirement paragraph	Inspection document
Treatment & painting	3.6	MIL-STD-171
Marking	3.7	MIL-M-13231
Workmanship	3.10	MIL-STD-252
Castings	3.8	MIL-C-6021 Class 2A
Solderless connectors	3.11	MIL-P-11268

4.3.2 Inprocess inspection not prescribed by a specification or drawing. Each lot of components and subassemblies not prescribed by a specification or a drawing with a specific inspection plan shall have all parameters examined for defects in accordance with the sampling plan tables of MIL-STD-105 with major defects (see 4.3.2.1) having an AQL of 1% defective combined and minor defects having an AQL of 4% defective combined. Inspection level II shall be applicable to both sampling plans.

4.3.2.1 Classification of defects for inprocess inspection not prescribed by a specification or drawing.

Major

101 Performance requirements specified in notes appearing on drawings.
102 Linear or diametral tolerances of 0.002 inch or less.
103 Angular dimensions having a tolerance of 15 minutes of arc or less.
104 Machine finishes of 32 micro-inches RMS or finer.
105 Tolerances of position or form 0.002 inch or less.
106 Screw thread Class 3 or better.
107 Load requirements for springs.
108 Lubrication requirements including dry lubricant finishes.
109 Material hardness when specified on drawings.

Minor

Defects not listed as major shall be classified as minor.

4.3.3 Examination. Each assembled searchlight shall be examined for defects in accordance with Table II. One or more defects shall be cause for rejection of that searchlight.

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TABLE II Examination

Inspection	Requirement paragraph	Inspection criteria
Any part or component missing or damaged	3.4	MIL-STD-252
Treatment and painting not as specified	3.6	MIL-STD-171
Markings not as specified or illegible	3.7	MIL-M-13231
Workmanship not as specified	3.10	MIL-STD-252

4.3.4 Tests. Using searchlights that have passed the examination of 4.3.3 the first article searchlights shall be subjected to the tests in Table III. Prior to start of tests, the searchlights shall be identified as to their applicable subgroups (see 6.2). Tests shall be conducted in the order listed. Failure of any test shall constitute first article failure.

TABLE III First Article Inspection

Inspection	Requirement paragraph	Test paragraph	Subgroups 1	2	3	4
Starting and normal operation	3.5.1	4.6.1	X	X	X	
Time to change operational mode	3.5.2	4.6.2	X	X	X	
Beam characteristics	3.5.6	4.6.6	X	X	X	
Light-tightness	3.5.4	4.6.4			X	
Angular visual security	3.5.5	4.6.5			X	
Mechanical shock	3.5.10	4.6.10			X	
Vibration	3.5.9	4.6.9			X	
High temperature storage	3.5.16.2	4.6.16.2			X	
High temperature operation	3.5.17.2	4.6.17.2			X	
Low temperature storage	3.5.16.1	4.6.16.1			X	
Low temperature operation	3.5.17.1	4.6.17.1			X	
Altitude	3.5.13	4.6.13			X	
Altitude transportation	3.5.14	4.6.14			X	
Salt fog	3.5.15	4.6.15			X	
Immersion	3.5.11	4.6.11			X	
Electromagnetic compatibility	3.5.18	4.6.18	X			
Angular visual security	3.5.5	4.6.5			X	
Beam characteristics	3.5.6	4.6.6			X	
RSI cooling	3.5.7	4.6.7			X	
Alinement retention	3.5.3	4.6.3			X	
Fungus and moisture resistance	3.5.12	4.6.12		X		
Starting and normal operation	3.5.1	4.6.1	X	X	X	
Time to change operational mode	3.5.2	4.6.2	X	X	X	
Beam characteristics	3.5.6	4.6.6	X	X	X	
Reliability qualification	3.5.8	4.6.8.3				X

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4.4 Inspections covered by subsidiary documents. This paragraph not applicable with this specification.

4.5 Quality conformance inspection.

4.5.1 Inprocess inspection. Inprocess inspection shall be as specified in 4.3.1 and 4.3.2.

4.5.2 Examination. Each assembled searchlight shall be examined as specified in Table II. Presence of one or more defects shall be cause for rejection of that searchlight.

4.5.3 Tests.

4.5.3.1 Group A Inspection. Following the examination specified in 4.5.2, each searchlight shall be tested in accordance with Table IV in the order listed. Failure of any test shall be cause for rejection of that unit.

TABLE IV Group A Inspection

Inspection	Requirement paragraph	Test paragraph
Starting and normal operation	3.5.1	4.6.1
Time to change operational mode	3.5.2	4.6.2
Beam characteristics	3.5.6	4.6.6
Light tightness	3.5.4	4.6.4

4.5.3.2 Group B Inspection. Group B inspection shall be conducted on searchlights selected from units which have passed the tests in 4.5.3.1. Sampling of lots (see 6.3) shall be in accordance with MIL-STD-105, AQL 2.5 percent defective, inspection level S-3 combined. Tests shall be conducted in accordance with Table V and shall be performed in the order listed.

TABLE V Group B Inspection

Inspection	Requirement paragraph	Test paragraph
Vibration	3.5.9	4.6.9 (c only)
Immersion	3.5.11	4.6.11

4.5.3.3 Group C Inspection. Group C inspections shall be conducted on searchlights selected from units which have passed the tests in 4.5.3.1. The sample(s) shall be subjected to the inspections specified in Table VI. Samples shall be selected in accordance with 4.5.3.3.1. Tests shall be performed in the order listed.

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4.5.3.3.1 Sampling for Group C Inspection.

4.5.3.3.1.1 Subgroup 1. For this subgroup 1 searchlight shall be selected from the first quality conformance inspection production lot. For subsequent Group C inspection, 1 searchlight shall be selected every month from the monthly specified quantity (see 6.2).

4.5.3.3.1.2 Subgroup 2. For this subgroup 1 searchlight shall be selected from the first quality conformance inspection production lot. For subsequent Group C inspection, 1 searchlight shall be selected every third month from the third monthly specified quantity, (see 6.2).

4.5.3.3.1.3 Subgroup 3. For this subgroup, 1 searchlight shall be selected from every sixth quality conformance inspection production lot.

TABLE VI Group C Inspection

Inspection	Requirement paragraph	Test paragraph
Subgroup 1		
Mechanical shock	3.5.10	4.6.10
Vibration	3.5.9	4.6.9 (a and b only)
Alinement retention	3.5.3	4.6.3
RSI cooling	3.5.7	4.6.7
Subgroup 2		
High temperature operation	3.5.17.2	4.6.17.2
Low temperature operation	3.5.17.1	4.6.17.1
Altitude	3.5.13	4.6.13
Angular visual security	3.5.5	4.6.5
Subgroup 3		
High temperature storage	3.5.16.2	4.6.16.2
Low temperature storage	3.5.16.1	4.6.16.1
Altitude transportation	3.5.14	4.6.14
Fungus and moisture resistance	3.5.12	4.6.12
EMI	3.5.18	4.6.18

4.5.3.3.2 Group C failures. Actions required relative to Group C failures shall be as specified in the contract or order, (see 6.2).

4.5.3.4 Group D Inspection. Group D inspection shall be conducted on searchlights selected from units which have passed the tests in 4.5.3.1. Tests shall be conducted in accordance with Table VII.

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4.5.3.4.1 Sampling for Group D inspections. The acceptance test shall be performed on the first production lot and every 3 months or 300 searchlights thereafter, whichever occurs first. Sample size shall be at the discretion of the contractor subject to:

a. Minimum sample size of 3.

b. Selection of sufficient samples to ensure completion of test prior to next sampling period.

TABLE VII Group D Inspection

Test	Requirement paragraph	Test paragraph
Reliability	3.5.8	4.6.8.4

4.5.3.4.2 Group D failures. Actions required relative to Group D failures shall be as specified in the contract or order, (see 6.2).

4.5.3.5 Reinspection of Group B, C, and D sample units. Unless otherwise specified, sample units which have been subjected to and passed Group B, C and D inspections may be accepted on contract or order provided all damage is repaired and the units are resubjected to and pass Group A inspection.

4.6 Test methods. Tests required to determine compliance with the requirements of this specification shall be conducted in accordance with the test procedures specified herein. Unless otherwise specified, input voltage shall be 28, plus 0, minus 1 vdc, voltages specified as 24 vdc shall be plus or minus 0.5 vdc, Celsius temperatures shall be plus or minus 2°C, time expressed in hours and/or minutes shall be plus or minus 5 minutes. All voltages and current measurements shall be made after 5 minutes minimum of searchlight operation. Searchlights having beam characteristics measured for the first time shall be operated for 2 hours prior to the measurements.

4.6.1 Starting and normal operation. Prior to initiating operation of the searchlight, the test selector switch on the control box shall be rotated through all of its positions with dc input voltage applied. The searchlight shall be ignited and ignition maintained at 24 volts dc input. The searchlight shall be turned "OFF" and reignited with an input of 28 volts dc. Failure of the searchlight to ignite, ignition time in excess of 3 seconds, input current in excess of 60 amperes, failure of the white indicator lamp on the control box to remain "OFF" the selector switch number 1 (operate) position and to be "ON" in all other selector switch positions shall constitute failure of the test.

Note: A decaying intensity of the indicator lamp is not a failure criteria.

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4.6.2 Time to change operational mode. The searchlight operating at 24 volts dc input shall be switched from each of compact visible, compact infrared, spread visible, and spread infrared modes to every other mode. The time shall be measured from the closing of the switch to completion of the operation. Any resultant mode change requiring more than three seconds shall constitute failure of this test.

4.6.3 Alinement retention. A point light source shall be projected to a flat reflecting surface attached to the front lamp clip assembly and reflected to a wall or flat surface that is not less than 15 feet away. The location of the environment on a shock testing machine for lightweight equipment as specified in MIL-S-901, Shockproof Equipment Class HI (High Impact) and in Method 207, MIL-STD-202. The test shall be conducted in 2 steps, as follows:

a. Two shocks shall be applied parallel to the optical axis of the searchlight with the hammer raised 12 inches from the anvil plate.

b. Two shocks shall be applied in a vertical direction perpendicular to the optical axis with the vertical hammer being dropped a distance of 6 inches from the anvil plate.

Tests 4.6.1 and 4.6.2 and 4.6.6 shall be performed. Failure to comply with 3.5.1, 3.5.2 and 3.5.6 or failure of the projected light source to return to the initial point within plus or minus 5 milliradians in any direction shall constitute failure of the tests.

4.6.4 Light tightness. The searchlight, in an area having an ambient illumination not greater than 10^{-5} footcandles, shall be operated at 28 vdc input in the infrared mode. Three observers having normal vision or corrected vision to 20/20 shall be used for this test. The observers shall be dark adapted for at least 30 minutes in the same test area that the test will be performed. The observers shall observe the searchlight for light leakage from the front, rear, and both sides at a distance of 25 feet with the following exception: The searchlight shall not be viewed within an angle of 5° from the beam axis. Detection of direct or reflected radiated white light at any angle at a distance of 25 feet shall constitute failure of this test.

4.6.5 Angular visual security. The searchlight shall be mounted on a goniometer. The test shall be conducted on an outdoor range at night with ambient conditions not to exceed 2×10^{-3} footcandles. The searchlight operating at 28 volts dc input and in the visible compact mode, shall have the beam projected on a flat target area 1000 meters from the searchlight. The center of the beam shall be not more than 5 feet or less than 4 feet from the ground plane at the target area. After placing the searchlight in the compact infrared mode, 3 observers, with vision corrected, if necessary, to 20/20 shall be dark adapted for 30 minutes prior to starting tests. The observers, one at a time shall be positioned at the target area facing the direction of the searchlight. The searchlight shall be moved slowly from center until the observer can not see the red glow and then shall be moved back through center to the opposite side until observer can no longer see the

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red glow. The two readings shall be recorded with each observer and averaged. Repeat this observation test 3 times. An average reading greater than 2-1/2° from center (total of 5°) shall constitute failure of this test.

4.6.6 Beam characteristics. During the inspection of beam characteristics the luminous intensity shall be measured with a photometer, corrected to have a response within plus or minus 5 percent of the standard international commission on illumination curve for photopic vision of the human eye. The photometer shall be calibrated monthly to relate meter readings to footcandles by comparison with the illuminance in footcandles of a laboratory standard lamp operating at a color temperature of 2870° Kelvin (K), plus or minus 50° K. With the searchlight mounted in a goniometer, the compact visible mode shall be activated. The searchlight beam shall be projected into a parabolic collimating mirror with a usable diameter larger than the full aperture of the searchlight. The collimating mirror shall focus the searchlight beam onto a circular collection aperture located at the Effective Focal Length (EFL) of the mirror. The diameter of the aperture divided by the EFL of collimating mirror shall be one-tenth or less than the diameter of the circle of confusion of the searchlight divided by the searchlight EFL. An optical relay system shall be positioned behind the collection aperture to transmit light into the photometer. The searchlight shall be rotated in the goniometer to produce data resulting in a graphical plot of intensity versus angle. The searchlight shall be placed into the spread visible mode and another intensity versus angle plot obtained. Luminous intensity in candelas shall be calculated from the footcandle values of the photometer by multiplying the number of footcandle values of the photometer by multiplying the number of footcandles by the square of the collimator effective focal length in feet and dividing by the collimator mirror reflectance. Failure to comply with 3.5.6 shall constitute failure of this test.

4.6.7 RSI cooling. The searchlight shall be equipped with thermocouples at the exhaust grill and the searchlight case to measure the exhaust temperature of the cooling air. The searchlight shall be operated in the infrared mode for a minimum of 15 minutes at 28 vdc input in an ambient temperature of plus 23°C, plus or minus 5°C. The maximum temperature of the exhaust air shall be recorded. The searchlight RSI switch shall then be turned to "OFF" position. Temperature limits in excess of those specified in 3.5.7 shall constitute failure of this test.

4.6.8 Reliability.

4.6.8.1 Operating characteristics. During reliability qualification and acceptance tests, each searchlight assembly shall be tested to insure compliance with the following requirements initially and at the end of every sequence of Figure 1.

a. Starting and normal operation (3.5.1)

b. Beam characteristics (3.5.6)

c. Time to change operational modes (3.5.2)

No single searchlight "ON" time shall be less than one half the average operating time of all equipment in each respective test.

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4.6.8.2 Accountable defects for reliability qualification and acceptance tests. An accountable defect is the inability of the searchlight to perform the operating characteristics tests of 4.6.8.1. Defects which are attributable to the RSI after 300 hours of "ON" time shall not be counted. In the event of a defect whose actual time of occurrence is unknown, the time of occurrence will be determined to be the end of the last complete sequence of Figure 1 at which the parameter in question was successfully measured.

4.6.8.3 Reliability qualification.

4.6.8.3.1 Test conditions. The qualification test shall be in accordance with Test Plan III of MIL-STD-781 using a specified MTBF of 500 hours. Test level E of MIL-STD-781 with the following exceptions shall be applied.

Temperature	Plus 15°C to minus 40°C
Temperature cycling	See Figure 1
Equipment ON-OFF cycling	See Figure 1
Input voltage cycling	Not required
Vibration	Total excursion of 0.06 inch at any non-resonant frequency between 10 and 55 Hz with a duration of at least 5 minutes during each hour of operating time.
Input voltage	28 vdc, plus 0 vdc, minus 1 vdc.

The searchlight with the main control box shall be so arranged in an environmental chamber that it may be operated from outside the chamber using the remote control box.

4.6.8.3.2 Operating characteristics. Operating characteristics shall be in accordance with 4.6.8.1.

4.6.8.3.3 Failure criteria. The accept/reject criteria shall be in accordance with MIL-STD-781 Test Plan III.

4.6.8.4 Reliability acceptance.

4.6.8.4.1 Test conditions. The acceptance test shall be in accordance with Test Plan III of MIL-STD-781 using a specified MTBF of 500 hours. Test level B of MIL-STD-781 with the following exceptions shall be applicable.

Temperature: Room Temperature (see 6.3.4)

Vibration: None

Temperature Cycling: None

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Sequence of Figure 1: 1.6 hours "ON", 0.4 hours "OFF" with each "ON" period cycled 0.8 hour visible mode-spread beam and 0.8 hour IR mode narrow beam.

Input voltage cycling: As specified in 4.6.8.3.1.

4.6.8.4.2 Operating characteristics. Operating characteristics shall be in accordance with 4.6.8.1.

4.6.8.4.3 Failure criteria. The accept/reject criteria shall be in accordance with MIL-STD-781, Test Plan III.

4.6.9 Vibration. The searchlight with the appropriate mounting kit (see 6.2) shall be rigidly mounted to a vibration table. Control boxes and cables may be mounted to the same vibration table or may be vibrated separately in any or all planes of vibration. Throughout the procedure listed below the searchlight shall be operated at an input voltage that will provide a RSI current of 50 to 54 amperes. The maximum input to the control box shall be 29.5 vdc. Within two minutes after initiation of searchlight operation, the procedures listed, below shall be started. Vibration amplitude shall be 0.03 inch (0.06 inch, plus or minus ten percent total excursion) from 10 to 55 Hz and 5 g's from 55 to 500 Hz. At the conclusion of the vibration, 4.6.1, 4.6.2, 4.6.4 and 4.6.6 shall be performed. Failure to comply with 3.5.1, 3.5.2, 3.5.4 or 3.5.6 or evidence of damage shall constitute failure of this test.

a. Vibrate the searchlight in a horizontal plane, with the optical axis parallel to the direction of vibration. Vibration frequency shall sweep logarithmically from 10 to 500 Hz and back to 10 Hz within 5 minutes for a total of five cycles. The resonant frequencies of the searchlight shall be recorded. Vibrate the searchlight at each of the resonant frequencies for five minutes. When no resonant frequency is observed, vibrate the searchlight at 26 Hz for five minutes.

b. Reapeat a. with the optical axis perpendicular to the direction of vibration.

c. Repeat a. with the vibration in the vertical plane while performing tests in 4.6.1 and 4.6.2. (These tests shall be performed at least once for each complete cycle.)

d. Repeat a. through c. at minus 54°C starting within five minutes after exposure to minus 54°C ambient temperature for 12 hours.

4.6.10 Mechanical shock. The searchlight operating with an input of 28 volts dc and mounted (see 6.2) shall be subjected to 4 shock impacts. The shock measurements shall be taken at the searchlight attach points. The test shall be conducted in 2 steps as follows:

a. Two shocks of 55 g's shall be applied parallel to the optical axis of the searchlight with the time duration of each to be 6.0 milliseconds, plus or minus 0.5 millisecond.

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b. Two shocks of 20 g's shall be applied in a vertical direction perpendicular to the optical axis of the searchlight with the time duration of each shock to be 6.0 milliseconds, plus or minus 0.5 millisecond.

Tests 4.6.1, 4.6.2, 4.6.4 and 4.6.6 shall be performed. Failure of the light to stay on or reignite within 3 seconds during the test, damage or failure to comply with 3.5.1, 3.5.2, 3.5.4 and 3.5.6 shall constitute failure of this test.

4.6.11 Immersion. The searchlight shall be immersed in fresh water to such a depth that no portion of the searchlight is less than 3 feet below the water surface. After 30 minutes minimum of immersion, the searchlight shall be removed from the water and examined for damage and leaks. Within 15 minutes after removal from the water, the searchlight shall be tested as specified in 4.6.1 Presence of water inside the housing, or failure of the test specified in 4.6.1 shall constitute failure of this test.

4.6.12 Fungus and moisture resistance. The searchlight shall be subjected to the test of Procedure I, Method 508 of MIL-STD-810. All gasket-sealed, or other nonhermetic-sealed equipment shall be sprayed with spore suspension and exposed. The heat exchanger and optical components (RSI, reflector, IR filter) shall not be sprayed. Operation of the equipment shall not be required. At the conclusion of the exposure period, examination shall reveal no evidence of visible fungus or corrosion of any equipment surfaces with the exception of those areas not sprayed.

4.6.13 Altitude. The searchlight shall be placed in the test chamber. The air pressure in the test chamber shall be reduced to the value corresponding to 10,000 feet above sea level. The temperature shall be controlled to plus 35°C, plus or minus 2°C. Th searchlight shall be ignited and operated continously for 1 hour at 28 vdc, 50 to 60 amperes. Failure of the searchlight to ignite and maintain ignition within 3 seconds and operate for 1 hour at 28 vdc, 50 to 60 amperes or evidence of damage, shall constitute failure of this test.

4.6.14 Altitude transportation. The searchlight shall be placed in a test chamber and the air pressure in the chamber shall be reduced to the value corresponding to 40,000 feet above sea level with an ambient air temperature of minus 62°C. The chamber shall be maintained at this pressure and temperature for 4 hours. At the end of 4 hours the chamber air pressure shall be returned to the value corresponding to sea level. The searchlight shall then be examined for damage and tested per 4.6.1. Evidence of damage or failure to comply to 3.5.1 shall constitute failure of this test.

4.6.15 Salt fog. The searchlight shall be subjected to Procedure 1 Method 509 of MIL-STD-810, followed by 4.6.6, beam characteristics. Evidence of damage, failure to comply with 3.5.6 or with MIL-STD-810 shall constitute failure of this test.

4.6.16 Storage temperatures.

4.6.16.1 Low temperature storage. The searchlight shall be placed in the test chamber and exposed to an ambient temperature of minus 62°C and maintained for a period of 12 hours. The ambient temperature shall be raised to room temperature

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and maintained for 6 hours. The searchlight then shall be tested per 4.6.1. Failure to comply with 3.5.1 or evidences of damage shall constitute failure of the test.

4.6.16.2 High temperature storage. The searchlight shall be placed in the test chamber and exposed to an ambient temperature of plus 68°C with maximum humidity (90 to 98 percent) for 12 hours. The ambient temperature shall be lowered to room temperature and maintained for 6 hours. After the exposure period, the searchlight shall then be tested per 4.6.1. Failure to comply with 3.5.1 or evidence of damage shall constitute failure of the test.

4.6.17 Operating temperatures.

4.6.17.1 Low temperature operation. The searchlight shall be exposed to an ambient temperature of minus 54°C for a period of 12 hours. At the end of the 12 hour period the searchlight shall be tested as specified in 4.6.1 and 4.6.2. Failure to comply with 3.5.1 and 3.5.2 shall constitute failure of this test.

4.6.17.2 High temperature operation. The searchlight shall be placed in a test chamber and subjected to one 12 hour cycle of exposure with a relative humidity maintained between 90 to 98 percent at all time. This cycle shall consist of the following:

a. Four hours at plus 52°C.

b. Decrease temperature within 1 hour to plus 23°C.

c. Maintain plus 23°C for 2 hours.

d. Increase temperature within 1 hour to plus 52°C.

e. Maintain plus 52°C for 4 hours.

The searchlight shall be operated during the third and fourth hours of the test cycle at plus 52°C and during the last 2 hours at plus 52°C after which test 4.6.1 shall be performed. Failure of the searchlight to ignite, failure to comply with 3.5.1 or evidence of damage shall constitute failure of this test.

4.6.18 Electromagnetic compatibility. The searchlight shall be tested in accordance with MIL-STD-462, Equipment Class II C Tactical Non-CE. Failure to comply with 3.5.18 shall constitute failure of the test.

4.7 Inspection of preparation for delivery. Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A, or C, as specified. (See 6.2)

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5.1.1 Level A.

5.1.1.1 Cleaning. Each Searchlight, Infrared, 28 Volt, 50-60 Ampere, AN/VSS-3, shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Each Searchlight, Infrared, 28 Volt, 50-60 Ampere, AN/VSS-3, shall be dried in accordance with the applicable precedure of MIL-P-116.

5.1.1.3 Preservation application. None required.

5.1.1.4 Unit packaging. Unit packaging shall be in accordance with the method prescribed in MIL-P-116, as specified herein.

5.1.1.4.1 Technical literature. Each technical literature shall be packaged method 1C-1.

5.1.1.4.2 Control, C-7905/VSS-3, Control, C-7906/VSS-3, Searchlight Mount, with Power Cable, Lamp Holder and Canvas Cover. Each item shall be individually packaged method III as follows: Cushion each item by wrapping in paperboard conforming to PPP-P-291, Type I. Secure wrapping with gummed paper tape conforming to PPP-T-45, Type III. grade A. Place each cushioned item within a clean-fitting box conforming to MIL-B-43014, Form I, Style II. Closure shall be in accordance with the appendix of the box specification.

5.1.1.4.3 Spare lamp and holder assembly. Each spare lamp and holder assembly shall be individually packaged method III as follows: Cushion the item on all surfaces with pads of polyurethane foam fabricated of material conforming to MIL-P-26514, designed to protect all projections and absorb the shock of impact in handling and transit. Place the cushioned lamp and holder assembly within a close-fitting box conforming to MIL-B-43014, Form 1, Style II. Closure shall be in accordance with the appendix of the box specification.

5.1.1.4.4 Hardware. Hardware shall be packaged method III as follows: Place all hardware within a cloth bag conforming to PPP-B-20 and secure ties.

5.1.1.4.5 Consolidated package. Consolidate the items packaged as specified in 5.1.1.4.2 thru 5.1.1.4.4, within a close-fitting box conforming to PPP-B-636, W5c. Fill all voids with paperboard conforming to PPP-P-291, type 1, to prevent movement. Closure shall be in accordance with the appendix of the box specification.

5.1.1.4.6 Searchlight MX-8272/VSS-3. Each Searchlight MX-8272/VSS-3, shall be individually packaged method III as follows: Cushion each searchlight on all surfaces with pads of polyurethane foam fabricated of material conforming to MIL-P-26514 designed to protect all projections and absorb the shock of impact in handling and transit. Place the cushioned item within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Closure shall be in accordance with the appendix of the box specification.

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5.1.2 Level C. Searchlight, Infrared, 28-Volt, 50-60 Ampere AN/VSS-3, shall be preserved and packaged in a manner that will afford adequate protection against physical and environmental damage during shipment, handling and limited intransit storage.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2). Shipping containers for all levels shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

5.2.1 Level A. Searchlight, Infrared, 28 Volt, 50-60 Ampere, AN/VSS-3, packaged as specified in 5.1 shall be individually packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or w, class 3. Fiberboard boxes being placed directly in the shipping container shall be waterproofed with tape conforming to PPP-T-76, in accordance with the taping requirements of the appendix of the box specification. Place the technical literature, packaged as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the box. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type 1, finish A.

5.2.2 Level B. Searchlight, Infrared 28 Volt, 50-60 Ampere, AN/VSS-3, packaged as specified in 5.1, shall be individually packed within a close-fitting box conforming to PPP-B-601, domestic type; PPP-B-621, style 4, class 1; or PPP-B-585, style 2 or 3, class 3. Place the technical literature packaged as specified in 5.1.1.4.1 on top of the contents, directly under the lid of the box. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto.

5.2.3 Level C. Searchlight, Infrared 28 Volt, 50-60 Ampere AN/VSS-3, packaged as specified in 5.1, shall be packed in shipping containers that comply with Federal or Military specifications, in a manner that will afford adequate protection to the package and its contents against physical and environmental deterioration and damage, during limited shipment, handling and intransit storage.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The AN/VSS-3 searchlight is intended for mounting on tanks or other vehicles with the capability of engaging targets at night. The visible beam illuminates targets for observation through normal daylight optical systems. The infrared beam illuminates targets for observation through the infrared optical systems while maintaining maximum security possible with the type of filter being used.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. When first article tests are to be conducted and the number of searchlights in each subgroup (see 3.3, 4.3, and Table III). It is recommended that one searchlight be used in each of subgroups 1 and 2, two searchlights in subgroup 3, and three searchlights be used in subgroup 4. The subgroup testing may be performed concurrently.

c. Time frame required for submission of first article test reports.

d. Time frame required for approval of the first article test reports by the Contracting Officer.

e. Production delivery schedule.

f. Level of preservation and packaging and the level of packing required (see 5.1 and 5.2).

g. Time interval required for submission of electromagnetic compatibility report.

h. The necessary mounting kit, either SC-DL-647180 (M60) or SC-DL-647132 (M551) required. (see 3.1).

i. Actions required relative to Group C and Group D failures. (see 4.5.3.3.2 and 4.5.3.4.2.)

j. When first article is required on MIL-F-45289 filters (see 3.4.2) and MIL-R-55344 Radiation Source (see 3.4.1) and the disposition of the first article Radiation Sources and Filters.

6.3 Definitions. For the purpose of this specification, the following definitions apply.

6.3.1 Compact mode (visible and infrared). Compact mode is the operating condition of the searchlight when the beam width is restricted to minimum.

6.3.2 Spread-beam mode (visible and infrared). Spread-beam is the operating condition of the searchlight when the beam width is expanded to maximum.

6.3.3 Damage. Damage is defined as deformation, loosening, breakage, corrosion, or change of fit of any component or part.

6.3.4 Room temperature. Room temperature is defined as plus 23°C, plus or minus 3°C.

6.3.5 "g". g is defined as an acceleration or deceleration of 32.17 feet per second per second.

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6.3.6 White light. The red glow seen when looking at the front of the searchlight (in the infrared mode) may contain pinholes of white light (when clear glass is used as filter substrate), which shall not be detectable at 26 feet or beyond. When performing test 4.6.4, the presence of this condition should not be considered as cause for rejection.

6.3.7 Lot size. A lot is defined as those searchlights successfully completing Group A test for any one month.

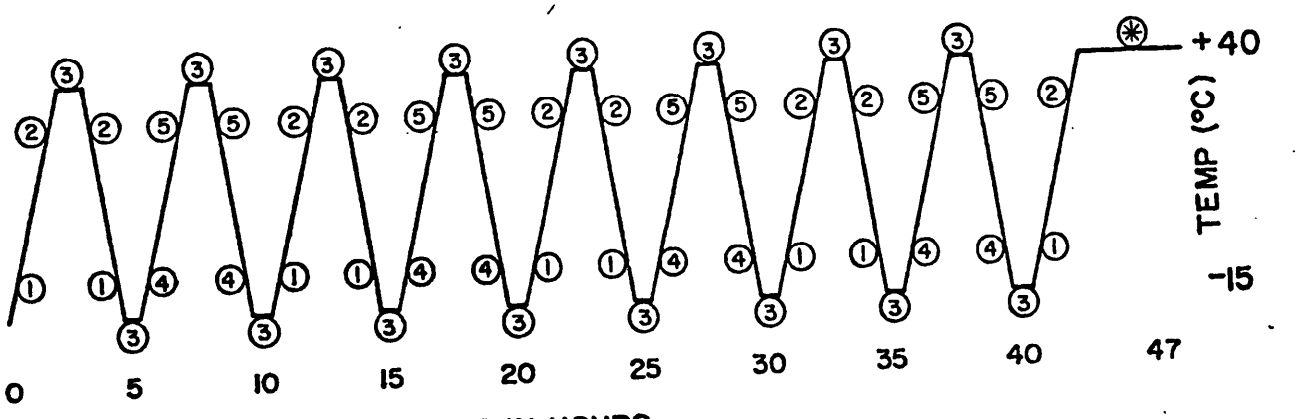
CUSTODIAN:
Army - EL
Navy - EC
Air Force -

USER ACTIVITY:
Army - MU

PREPARING ACTIVITY:
Army - EL

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*REMAIN IN IR COMPACT UNTIL THE 47TH HOUR.

MODE	TIME IN MODE	SYMBOL OF MODE
VIS. SPREAD	1 MINUTE	①
IR COMPACT	1 HOUR 59 MINUTES	②
OFF	30 MINUTES	③
IR SPREAD	1 MINUTE	④
VIS. COMPACT	1 HOUR 59 MINUTES	⑤

RELIABILITY SEQUENCE
FIGURE 1

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-S-55343B SEARCHLIGHT, INFRARED 28 VOLT, 50-60 AMPERE AN/VSS-3

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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